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Factors Affecting the Prevalence of Tinnitus

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Abstract. The prevalence of tinnitus was studied in a large population of noise-exposed workers. It was found that 6.6% had tinnitus which was more than momentary. Results of two other studies were compared. Factors possibly relating to the prevalence of tinnitus were studied. The single most important factor found to be related to tinnitus is hearing level. The higher the hearing level, the faster the rate of increase in the prevalence of tinnitus. Other factors such as sex, age, laterality, smoking and shooting do not seem to have a significant, direct relationship with tinnitus, but they are related to the prevalence of tinnitus indirectly through the influence they have on hearing loss.

Introduction

There are various epidemiological studies done on the characteristics and prevalence of tinnitus and hearing loss both in the United States [Singer et al., 1982] and the United Kingdom [Coles et al., 1981]. The percentage of people with tinnitus found differs considerably among studies. A major factor in the difference is the criteria by which the tinnitus cases were selected. If the study was done using questionnaires then how the questions were asked would have a great influence on the prevalence rate. Often cases with spontaneous, momentary tinnitus were included in these studies. There are factors which affect the probability of having tinnitus. These factors, therefore, in turn may affect the prevalence rate if sampling is not random or controlled for these factors. In this study some of the possible factors which might affect the probability of having tinnitus were examined in a large noise-exposed population. The factors included are sex, age, hearing loss, smoking, and shooting.

Method

In British Columbia, the Workers' Compensation Board (WCB) has jurisdiction over hearing conservation programs in industry. It is required by the WCB that all workers exposed to a time-weighted average of 85 dB A or more for 8 h shall have a pure-tone air conduction audiogram taken annually by a technician trained by the WCB. The audiometric data of each worker, along with the information on medical history, shooting history, smoking history, and the use of hearing protection are completed on an optical scan form by the technicians. The information is stored directly on tape for future utilization and analysis. Part of the optical scan form relevant to this study is replicated in the 'Appendix'. One of the questions in the section 'Medical history (hearing)' is: 'Do you have ringing in your ears?' The answer may be yes or no, and if the answer is yes, then left and/or right ear has to be indicated. The proper method of interpreting and completing the optical scan form is part of the 3-day intensive training and the 1-day refresher course at the WCB for the technicians. The question, 'Do you have ringing in your ears?' should be interpreted as tinnitus that is present more than just momentarily and at least recurring if not continuous.

Approximately 30 000 cases were analysed in this study. All these cases answered no to questions: 'Have you ever had (A, B, C, D, E, F, G, and J)?' and 'Do you now have (C and D)?' in the 'Medical history (hearing)' section in the 'Appendix'. This was done to eliminate as many as possible of the hearing-impaired cases which have other origins than noise damage. The data were analysed to answer the following questions: (1) Is tinnitus related to gender? (2) Does the prevalence of tinnitus vary with age? (3) How does hearing loss relate to tinnitus? (4) Does unilateral tinnitus usually occur in the ear with worse hearing? (5) Is there a difference in the prevalence of tinnitus between the left and the right ear? (6) Does smoking or shooting relate to the prevalence of tinnitus?

Results

The prevalence of unilateral and bilateral tinnitus in our population was found to be 6.6%. This number seems to vary substantially when groups with different age, hearing loss, etc. are considered. The following describes the prevalence of tinnitus when these factors are taken into account.

Prevalence of Tinnitus by Gender

A number of workers with unilateral or bilateral tinnitus was found in both the male and female population. Unfortunately, female workers only constituted about 5% of the total workers analysed. There were, however, still 1 664 female workers. Table I shows the 2 × 2 contingency table of sex versus tinnitus. The chi square test shows that there is a significant higher prevalence of tinnitus in the male population. However, male workers are known to have much worse mean hearing thresholds than

Table I. Prevalence of tinnitus: male versus female

	Male		Female		Male and female	
	n	%	n	%	n	%
Tinnitus	2 081	6.6	93	5.6	2 174	6.6
No tinnitus	29 423	93.4	1 571	94.4	30 994	93.4
Total	31 504		1 664		33 168	

Table II. Prevalence of tinnitus by age

	Age						
	< 19	20–29	30–39	40–49	50–59	60–69	> 70
% reported tinnitus	3.1	4.7	5.1	6.4	10.2	12.1	20.0
Total number in each group	94	8 376	8 413	5 804	4 723	1 994	16

female workers. It is possible that the higher prevalence of tinnitus in male workers is the consequence of their higher hearing threshold. The data were analysed for the sex factor in groups with similar hearing threshold levels. Those with hearing levels within the same decade of decibel hearing loss averaged across 3, 4, and 6 kHz were grouped together. When hearing threshold level was controlled in this manner there was no significant difference in the prevalence between the two sexes.

Prevalence of Tinnitus by Age

The prevalence of tinnitus by age was only studied for male workers because (1) it was shown above that there is no significant difference due to sex and (2) the sample size of workers in the older female groups was so small that they were not likely to yield true prevalence rates. Table II shows the prevalence of tinnitus by age in decades for males only. Again both unilateral and bilateral tinnitus were considered together. The prevalence can be seen to increase with age. However, increase in age generally also means increase in noise exposure and hearing loss in this population.

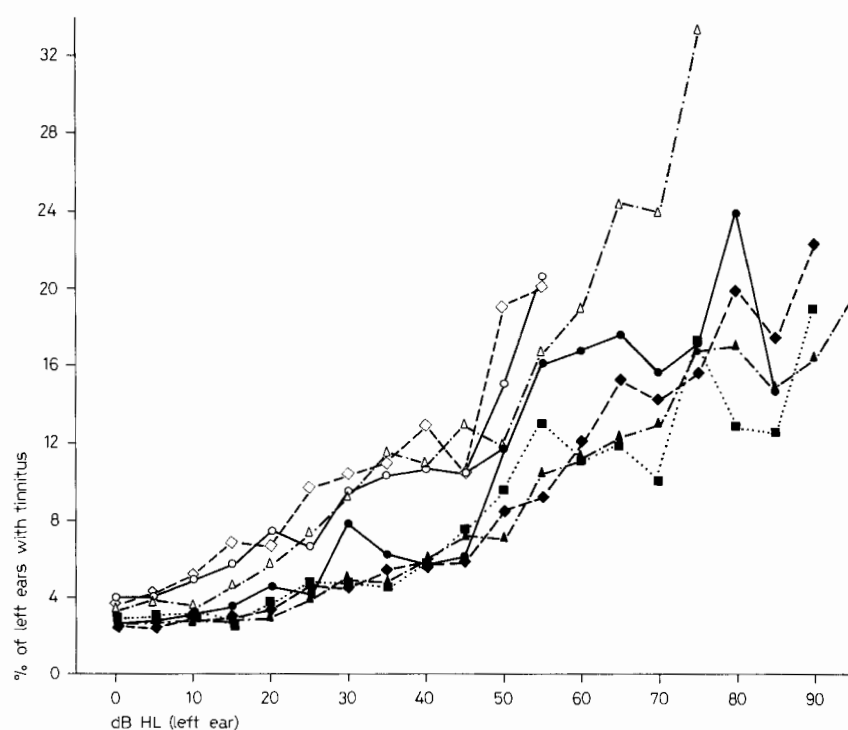


Fig. 1. Percentage of left ears with tinnitus versus hearing loss in the left ear in decibels HL for the male workers. Each symbol represents one frequency. ○ = 500 Hz; ◇ = 1 kHz; △ = 2 kHz; ● = 3 kHz; ◆ = 4 kHz; ▲ = 6 kHz; ■ = 8 kHz.

It is entirely possible that tinnitus is really related to hearing loss rather than age per se. In the next section on tinnitus by hearing loss, it can be seen that this is, in fact, the case.

Prevalence of Tinnitus as a Function of Hearing Loss by Frequency

In an epidemiological study, Coles et al. [1981] have shown that groups having tinnitus had much worse hearing than the comparable non-tinnitus groups. In this study, because of the availability of the large sample size the prevalence of tinnitus as a function of hearing loss at 5-dB intervals by frequency can be plotted as shown in figure 1. Only results of the left ear are shown, but the functions are very similar for both ears. Each datum

point represents a sample size of at least 50 workers, but about 80% of them represents a sample size of more than 300.

There is a distinct increase in the prevalence of tinnitus as hearing loss increases. The fact that the functions were plotted for all frequencies tested does not imply that they are independent of one another. Since by far the majority of hearing loss is noise-induced, high-frequency sensorineural type, the low-frequency functions (0.5, 1, and 2 kHz) do not likely reflect the true relationship between tinnitus and hearing loss. Figure 1 shows that the tinnitus versus hearing loss functions of 0.5, 1, and 2 kHz are very similar among themselves, but are higher than the high-frequency functions. This is expected because for the entire population the hearing level averaged across 3, 4, and 6 kHz is about 13 dB higher than that averaged across 0.5, 1, and 2 kHz.

If the high-frequency functions in figure 1 represent the relationship between tinnitus and hearing loss one can say that there is a positive relationship between the two ($p < 0.0001$) and that the relationship is non-linear, i.e. the higher the hearing loss, the faster the rate of increase in tinnitus cases. The spread of data is larger for higher than for lower hearing levels. This is at least partly due to the decrease in sample size as hearing level increases.

Inasmuch as hearing loss is positively related to age it is necessary to separate these factors. Figure 2 depicts the functions tinnitus versus hearing threshold level at 4 kHz for three age groups: 40–49, 50–59, and 60–69. Similar results are found for other frequencies, although they are not shown here. Only these three age groups were analysed because the sample size for higher hearing level was larger in these three groups. Each datum point represents a sample size of no less than 50. It can be seen in figure 2 that each of the functions tinnitus versus hearing level remains similar to that for 6 kHz in figure 1 even when controlled for age. The functions for the three age groups resemble one another closely. There is no apparent effect of age on tinnitus, at least for this age range.

Unilateral Tinnitus and Hearing Loss

It has been shown above that the greater the hearing loss, the higher the probability of having tinnitus. Unilateral tinnitus, therefore, is likely to be in the ear with more hearing loss. Table III shows the distribution of ears with the worse hearing and the side with tinnitus. The worse ear was defined in two ways, (1) when the hearing level averaged across 0.5, 1, and 2 kHz was worse and, (2) when averaged across 3, 4, and 6 kHz was worse.

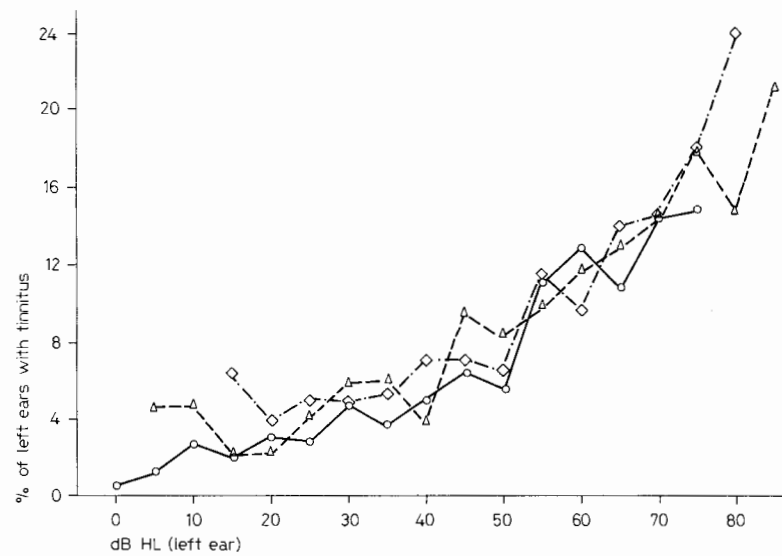


Fig. 2. Percentage of left ears with tinnitus versus hearing loss at 4 kHz in the left ear in decibels HL for male workers at age 40-49 (○), 50-59 (◇), and 60-69 (△).

Table III. The distribution of ears with the worse hearing and the side with tinnitus

Tinnitus	Worse ear at high frequencies		Worse ear at low frequencies	
	left	right	left	right
Left only	161	80	153	66
right only	150	103	92	139

The chi square test shows that the worse ear is significantly ($p < 0.01$) related to the ear with tinnitus by both definitions.

Prevalence of Tinnitus, Right versus Left

Table IV shows the number and percentage of male workers with tinnitus in the left ear and/or right ear. When only unilateral tinnitus was considered 0.8% had tinnitus in the left ear and 0.9% in the right ear. When both unilateral and bilateral tinnitus cases were considered together, 4.9% of the workers had tinnitus in the left ear and 5.0% in the right ear.

Table IV. The number and percentage of male workers with tinnitus in the left ear and/or the right ear

	Left	Right	Left only	Right only
Number	1 446	1 446	237	257
Percentage	4.9	5.0	0.8	0.9

Smoking and Shooting versus Tinnitus

Smoking history and shooting history were correlated with tinnitus. The chi square test showed that workers with a smoking history, either at present or in the past, have a significantly higher prevalence rate of tinnitus ($p < 0.0001$). This is also true for workers with a shooting history ($p < 0.02$). In a previous study [Chung et al., 1982] hearing loss was shown to be related to smoking. Workers with a history of smoking tend to have more hearing loss than workers without a history of smoking. It is, therefore, possible that tinnitus is only related to the smoking history as a consequence of its relation with hearing loss which is shown above to be related to tinnitus.

The relationship between tinnitus and smoking history was then tested controlling for hearing loss. This was achieved by grouping the workers in ages within the same decade and testing the relationship between smoking history and tinnitus. Similarly, the relationship between shooting history and tinnitus was tested. Both smoking and shooting history were not found to be significantly related to tinnitus in any of the age groups.

Discussion

There seems to be a marked difference in the prevalence of tinnitus when studies from the UK, the US and the present one are compared. In the UK study [Coles et al., 1981] approximately 17% of the 5 000 questionnaires studied replied having tinnitus. A recent study done in the US [Singer et al., 1982] on a sample of 6 913 adults found that 26.1% reported 'experiencing ringing at some point in the past few years', and when queried further only 8.6% reported tinnitus which occurred more than

momentarily. In the present study only 6.6% of a noise-exposed population reported tinnitus.

Demographic factors are not likely a major reason for the difference in results among the three studies. The two plausible key factors are:

(1) *The Method of Data Collection.* Coles et al. [1981] used questionnaires mailed out to a random sample. This could have led to a biased sample of those who responded. Furthermore, the interpretation of tinnitus depended substantially on the person completing the questionnaire. This type of survey might lead to an overestimation of the prevalence. This type of situation is clearly demonstrated in the US study in which only one-third of the original 26.1% who reported having ringing in the ear during the past few years actually had tinnitus which was more than momentary when further questioned.

(2) *The Distribution of Age or Hearing Loss in the Sample.* The probability of having tinnitus depends heavily on the hearing level. Naturally, the distribution of hearing level across the sample population is a key factor in determining the prevalence of tinnitus. Since age is a fair indicator of hearing level for a large group it is used here for comparing the three studies. Among the three studies, the present study is one which has the highest number of subjects under 40 years old (55%). For both of the other studies it was below 40%. It is not surprising that this study yields the lowest mean prevalence of tinnitus among the three.

The results of this study allow us to answer the five questions we have asked. In this discussion section, the questions and answers will be discussed one by one.

(1) *Is Tinnitus Related to Gender?* The answer is no. Results in the present study demonstrate no difference between the two groups when hearing threshold level is controlled. This supports the findings of Hazell [1981] in his 141 patients who reported tinnitus. There was approximately the same number of patients of each sex. Also, in the questionnaire survey done in the UK [Coles et al., 1981] no significant relationship between tinnitus and gender was found although a marginally higher percentage of women reported tinnitus than men (17.6 versus 16.9%). Generally, however, men have worse hearing than women. It should not be surprising then, if more men are found to have tinnitus than women when hearing level is not taken into account.

(2) *Does the Prevalence Rate of Tinnitus Vary with Age?* The answer is yes, simply because age is associated with hearing loss. Age, per se, is not a factor in tinnitus as can be seen in figure 2. Coles et al. [1981] in an extensive survey with questionnaires also found an increased percentage of people reporting tinnitus with age. Hearing loss, however, was not controlled in that study.

(3) *How Does Hearing Loss Relate to Tinnitus?* Hearing threshold seems to be the single most important factor that affects the prevalence of tinnitus. Results have indicated an exponential increase of tinnitus cases as hearing threshold increases. This same function may not apply to other types of hearing loss since the population in the present study is basically noise-exposed. Tinnitus apparently can occur without hearing impairment. The percentage of cases with tinnitus that are not associated with hearing loss varies with the definition of hearing impairment. If we take 25 dB hearing loss (ANSI, 1969) as the level above which hearing impairment is labelled then about 3% of the workers with no hearing loss at any frequency reported tinnitus. Among this 3% of workers there might be a small number who actually only had momentary tinnitus and reported having tinnitus. The 3% is small when compared with the percentage (~ 9%) found in the recent study conducted in the UK. However, the ~ 9% represents people who reported having tinnitus but no hearing impairment. It is likely that a number of people who reported no subjective hearing impairment, in fact, have hearing threshold above 25 dB HL at one or more frequencies. This may account for part of the difference between the two studies.

(4) *Does Unilateral Tinnitus Usually Occur in the Ear with Worse Hearing?* The answer is yes generally, but there are many considerations. In many situations it is difficult to define which ear has the worse hearing. By averaging the low or the high frequencies, as in this study, the thresholds of individual frequencies are lost. Perhaps that is the reason why the distribution in table III, albeit statistically significant, is not one which could be described as striking.

(5) *Is There a Difference in the Prevalence of Tinnitus between Left/Right Ear?* Contrary to expectation and other studies [Coles et al., 1981; Hazell, 1981] no lateral difference in the prevalence of tinnitus was found. In a previous study [Chung et al., 1983] the left ear has been found,

on the average, to have worse hearing than the right ear. The difference however, amounts to only a few decibels. Since the prevalence of tinnitus increases with hearing loss, one would expect a higher prevalence of tinnitus in the left ear. Table IV shows no indication of any lateral preference in tinnitus.

The apparent inconsistency of the lateral difference in hearing loss and the lack of lateral difference in the prevalence of tinnitus may be just because the difference of hearing loss between the two ears is not sufficient to cause an effect on the prevalence of tinnitus, and not due to other unknown factors.

(6) Does Smoking or Shooting Relate to the Prevalence of Tinnitus?

The answer is yes, but only indirectly. Both smoking [Chung et al., 1982] and shooting [Chung et al., 1981] have been shown to be related to hearing loss. It is not surprising that both smoking and shooting were found to be related to tinnitus. The finding that when controlling for hearing loss, neither smoking nor shooting was to be related to tinnitus, confirms the indirect effect of smoking and shooting on tinnitus.

Conclusion

The prevalence of tinnitus in a noise-exposed population has been found to be only 6.6%, a fairly conservative number when compared with two other survey studies. The single most important factor that was found to be related to tinnitus is hearing level. The higher the hearing level, the faster the rate of increase in the prevalence of tinnitus. Other factors such as sex, age, laterality, smoking and shooting do not have a significant direct relationship with tinnitus. However, they are related to the prevalence indirectly through the influence they have on hearing loss.

Facteurs affectant la prévalence des acouphènes

La prévalence des acouphènes a été étudiée dans une large population de travailleurs exposés aux bruits industriels. On a relevé que 6,6% des travailleurs avaient des acouphènes plus que passagers. Les résultats de deux autres recherches ont été comparés à celle-ci. Les facteurs associés à la prévalence des acouphènes ont été étudiés et seul le degré de surdité a

paru lié. Une surdité importante augmente la probabilité d'acouphènes chez les travailleurs. D'autres facteurs comme le sexe, l'âge, la latéralité, le fait de fumer ou de tirer ne semblent pas avoir de rapport direct ou significatif avec les acouphènes mais peuvent être liés à ceux-ci indirectement à travers leur influence sur la surdité.

Appendix

	No	Yes	Left	Right
<i>Medical history (hearing)</i>				
Have you ever had:				
A) A hearing test by an ear specialist in the last 5 years	_____	_____	_____	_____
B) A severe ear infection	_____	_____	_____	_____
C) Ear drainage or pain	_____	_____	_____	_____
D) Pressure in or around ears except altitude changes	_____	_____	_____	_____
E) Ear surgery	_____	_____	_____	_____
F) Dizziness or balance problems	_____	_____	_____	_____
G) A serious head injury	_____	_____	_____	_____
H) Exposure to loud blast or explosion	_____	_____	_____	_____
I) A hearing aid now or in the past	_____	_____	_____	_____
J) A relative with hearing loss before age 50	_____	_____	_____	_____
Do you now have:				
A) Trouble understanding speech	_____	_____	_____	_____
B) Ringing in your ears	_____	_____	_____	_____
C) Changes in hearing from day to day	_____	_____	_____	_____
D) A ruptured ear drum	_____	_____	_____	_____
<i>Shooting history</i>				
A) Do you/have you hunted	_____	_____		
B) Shot trap/skeet/target (not hand guns)	_____	_____		
C) Shot hand guns	_____	_____		
D) Worn ear protectors while shooting	_____	_____		
E) Shoulder shot from	L_____	R_____		
F) Number of years	< 10_____	10-20_____	21+_____	
<i>Smoking history</i>				
A) Do you smoke cigarettes now	_____	_____		
B) Did you in the past	_____	_____		
C) For how many years have you smoked	< 10_____	10-20_____	21+_____	
D) How many cigarettes per day do/did you smoke	< 10_____	10-20_____	21+_____	

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